

To Anchor or to Deliberate? AI Advice Timing in Cost-Sensitive Decision Support

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Abstract—Predictive models used in decision support are often evaluated as standalone classifiers, but deployment outcomes depend on how people use model outputs under task constraints. This paper tests whether AI-advice timing changes human–AI decision quality in a cost-sensitive loan task. A web-based within-subject experiment compared three protocols—no AI, AI-first, and human-first—using a calibrated logistic regression model held fixed across conditions. One hundred participants completed 1,800 scored trials under asymmetric penalties: approving a defaulter cost five units and rejecting a creditworthy applicant cost one unit. Both AI-assisted protocols significantly reduced mean trial cost relative to no AI: AI-first by 0.210 units ($p = 0.017$, $d = 0.243$) and human-first by 0.298 units ($p < 0.001$, $d = 0.385$). The direct timing contrast was not statistically decisive ($\Delta = 0.088$, $p = 0.178$, $d = 0.136$). In human-first trials, AI exposure corrected 90 initial errors versus 32 it introduced. Weight-of-advice analysis showed polarized reliance: 52.5% of trials showed no probability adjustment, while adjusted trials moved strongly toward the AI estimate (median WOA = 0.869). Although the sample was predominantly young non-experts in an 18-trial session, limiting external validity to professional settings, the findings support a system-level view: realized decision value depends not only on model quality but also on interaction structure, cost framing, and selective revision behavior.

Index Terms—human–AI interaction, decision support systems, advice timing, reliance, cost-sensitive evaluation, behavioral experiment

I. INTRODUCTION

AI decision-support systems are often justified using offline model metrics such as accuracy, AUC, or calibration error. These measures are necessary but not sufficient for evaluating deployed systems. In many decision contexts the model does not act autonomously; it advises a human decision-maker. The final decision is therefore produced by an interaction between the model output, the user, the interface, and the task incentives. A model can be statistically valid yet operationally ineffective if users ignore correct advice, over-follow incorrect advice, or apply the advice using the wrong decision threshold.

This paper studies one interaction-design choice that is often treated as a single unexamined default: *when* should AI advice be shown? In an AI-first workflow, the model output appears before the user forms a judgment. This may improve decision quality by supplying calibrated information early, but it also risks anchoring the user on the model output and reducing independent evaluation [1]. In a human-first

workflow, the user first forms an initial judgment and only then sees the AI recommendation. This structure can function as a cognitive forcing mechanism [2], preserving independent reasoning before advice exposure, but it may also create inertia that makes users less willing to adjust when the model is correct.

The question matters because prior work has studied algorithmic advice, overreliance, and cognitive forcing, but has less often isolated advice timing itself. A probability shown before judgment becomes an initial reference point; the same probability shown after judgment becomes advice that can be accepted, ignored, or used to revise. Thus, AI-first and human-first are distinct interaction structures even when the model and displayed information are held constant.

The study addresses three research questions. RQ1 asks whether AI probability support improves cost-sensitive decision quality relative to unaided human judgment. RQ2 asks whether advice timing changes realized decision quality when the model, cases, and information content are held constant. RQ3 asks whether, in the human-first workflow, AI exposure corrects more initial errors than it introduces and how strongly users move toward the AI estimate.

We evaluate advice timing in a loan approval task that exemplifies the human-centered AI research agenda [3]. Participants reviewed loan application profiles and made binary approve/reject decisions under asymmetric costs reflecting the fact that approving a defaulting loan is substantially more costly than rejecting a creditworthy borrower [4]. The AI system displayed a predicted probability of default derived from a calibrated logistic regression model. The domain provides a structured environment for studying human–AI decision support; no claims are made about professional lending practice.

The study makes three contributions. First, it provides empirical evidence from a completed within-subject experiment comparing no AI, AI-first, and human-first protocols while holding the predictive model and case set constant across conditions. Second, it evaluates decisions using a cost-sensitive benchmark rather than raw accuracy, exposing cost-relevant behavioral effects that accuracy metrics conceal. Third, the human-first protocol enables direct measurement of revision behavior, including whether AI exposure corrects initial errors or introduces new ones.

The main finding is that AI support improved decisions relative to no AI under both cost and accuracy metrics with medium effect sizes. Human-first support produced the lowest cost descriptively and showed clear correction behavior, but the direct timing difference versus AI-first was not statistically decisive at the achieved sample size.

II. LITERATURE REVIEW

A. The Model–System Performance Gap

A foundational challenge motivating this work is that deploying an accurate predictive model does not automatically improve decisions. Kleinberg et al. [5] demonstrated this through a retrospective policy simulation of bail decisions, showing via counterfactual analysis that an algorithm substantially outperformed judges at predicting flight risk. Their analysis was conducted on historical data rather than in a live human-AI interaction setting, leaving open whether such gains would materialize when decision-makers actually use algorithmic predictions.

Green and Chen [6] addressed this question directly through a controlled behavioral experiment. They found that algorithmic risk scores reliably improved participant accuracy over unaided baselines in most conditions, but participants failed to match the model’s standalone accuracy, could not evaluate prediction quality, and did not calibrate their reliance based on the model’s actual performance. We adopt their algorithm-in-the-loop framework as our central lens, treating the gap between model accuracy and realized decision quality as the primary quantity of interest.

Mozannar and Sontag [7] approached this gap from a machine learning perspective, formalizing the learning-to-defer problem: given that a human will make the final decision, how should the model be trained to allocate decision authority optimally? Their framework treats human performance as a function of case difficulty and formalizes the system-level objective that this paper studies empirically. We complement their theoretical contribution with behavioral data on how interaction structure, rather than model architecture, determines whether the potential for complementary performance is realized.

B. Reliance, Overreliance, and Cognitive Forcing

The human factors literature documents the tendency toward automation bias: the uncritical acceptance of automated recommendations even when human judgment would be superior [8]. In AI-assisted decision settings this manifests as overreliance—following the model when it is incorrect—which is systematically documented across clinical, judicial, and financial assessment domains [2], [9].

Buçinca et al. [2] showed that cognitive forcing functions—design interventions requiring users to form an independent judgment before receiving AI advice—can substantially reduce overreliance. Their finding is directly relevant to our human-first protocol, which embeds a structural cognitive forcing mechanism by collecting an initial decision before revealing

the AI prediction. Their work, however, does not systematically vary the timing of AI advice across a common case set while controlling for model quality and case difficulty, which is the precise contribution of this study.

Schemmer et al. [9] proposed the Appropriateness of Reliance (AoR) framework, decomposing reliance into two dimensions: correct AI reliance (following the AI when it is correct) and correct self-reliance (overriding the AI when it is wrong). This two-dimensional view is essential: a participant who always follows the AI scores high on the first dimension but near zero on the second. We operationalize a related decomposition in our human-first correction analysis, distinguishing beneficial reliance, harmful override, over-reliance, and beneficial override across the full trial set.

C. Timing, Anchoring, and Advice Taking

Advice timing is connected to anchoring. If the AI output is shown first, it may become the reference point around which the human forms a judgment [1]. Rastogi et al. [1] showed mathematically that anchoring on AI predictions can degrade collaborative performance and proposed enforced time delays as a de-anchoring mechanism. Their framework informs our measurement approach, though we manipulate the ordering of information rather than enforcing delays.

Steyvers and Kumar [3] identify sequential interaction and advice timing as important unresolved challenges in AI-assisted decision making, noting that the time-delayed paradigm has shown promise for reducing anchoring but that ordering effects remain understudied in fully within-subject designs. Bansal et al. [10] showed that complementary human–AI team performance depends on task structure and the conditions under which humans choose to override the model, suggesting that interaction design mediates whether potential complementarity is realized.

The weight of advice (WOA) metric [11], [12] measures the fraction of movement from an initial estimate toward an advisor’s estimate and is only observable when an initial judgment is collected before advice exposure. This makes WOA especially suited to the human-first condition. Prior WOA studies typically find conservative advice-taking with participants adjusting less than the normatively optimal amount, but the direction and magnitude of adjustment can vary substantially with advisor type and task domain.

D. Confidence, Calibration, and Uncertainty Communication

How AI uncertainty is communicated affects both reliance behavior and participant calibration. Zhang et al. [13] demonstrated that displaying model confidence alongside predictions can improve trust calibration—participants become more discriminating about when to follow AI advice—but that this benefit depends on the confidence signal being genuinely informative. When confidence is poorly calibrated, displaying it can increase overreliance.

This motivates our decision to use a probability-based AI representation derived from isotonic calibration [14] rather than a binary recommendation alone. Our system displays

both a predicted default probability and a derived cost-sensitive recommendation, allowing us to distinguish binary compliance from probability anchoring. The model’s mean confidence (probability assigned to the predicted class) averages 0.817 when correct and 0.731 when incorrect, yielding a discriminative gap that is large enough to be informative but small enough to create a genuinely non-degenerate reliance environment.

E. Case Difficulty and Complementary Performance

The relationship between task difficulty and human-AI team performance is central to achieving complementary outcomes. Bansal et al. [10] found that while AI augmentation produced complementary improvements over unaided human performance, AI explanations did not increase these gains and sometimes increased inappropriate reliance. A key implication is that complementary performance requires the human to selectively override incorrect AI predictions, which demands that human and AI errors be non-overlapping.

Our experimental design tests this directly. We construct 18 cases across three difficulty tiers (easy, medium, hard) and two model correctness conditions, with each block containing cases where the model is both correct and incorrect. This creates opportunities for both beneficial reliance and beneficial override, enabling measurement of whether interaction timing affects their balance. The medium-difficulty tier is of particular interest because cases in this range fall closest to the cost-sensitive decision threshold, where model-optimal disagreement is most consequential.

Credit decisions provide a well-structured domain for cost-sensitive human-AI research [4]. The cost asymmetry between false negatives (approving defaulters) and false positives (rejecting creditworthy borrowers) creates a normative framework that is absent in accuracy-only evaluation [15], enabling direct comparison between human behavior and a cost-minimizing benchmark.

III. METHODOLOGY

A. Prediction Task and Data

The loan default prediction task was built on LendingClub consumer loan data. The prediction target was whether a loan defaulted. Only variables observable at loan origination were used; post-outcome fields, payment-history variables, and internal credit-grade labels were excluded to prevent data leakage. The cleaned modeling dataset contained 1,298,960 loans issued from 2007 to 2018 with a global default rate of 20.1%.

The data were split temporally. Loans issued before 2016 were used for model development and calibration (default rate 18.5%); loans issued from 2016 onward formed the held-out test period from which experimental cases were drawn ($n = 479,744$; default rate 22.9%). This split preserves realistic temporal distribution shift and avoids using future lending outcomes to build the AI model shown to participants.

The final feature set contains nine variables satisfying four criteria: available at decision time, interpretable to lay

TABLE I
HELD-OUT MODEL PERFORMANCE ON THE POST-2016 TEST SET
($n = 479,744$)

Model	AUC	Brier	ECE
Logistic regression	0.691	0.164	0.033
XGBoost	0.697	0.163	0.034
Histogram gradient boosting	0.697	0.163	0.033
Bootstrap 95% CI for Logistic AUC: [0.690, 0.693]; Brier: [0.163, 0.165].			

participants, showing a monotonic relationship with default risk in training data, and non-redundant. The retained features are loan amount, term, interest rate, annual income (log-transformed), debt-to-income ratio, revolving utilization, home ownership, loan purpose, and credit history length. Redundant variables such as installment (functionally determined by loan amount, term, and rate) and low-signal variables such as delinquency count were excluded. Numeric features use median imputation and z-score standardization; categorical features use mode imputation and one-hot encoding.

B. AI Model

Three candidate classifiers were evaluated on a shared pipeline: logistic regression (L_2 regularization, $C = 0.1$), XGBoost (300 estimators, depth 4), and histogram gradient boosting (300 iterations, depth 4). Hyperparameters were fixed rather than grid-searched, as the objective is a stable, defensible probabilistic advisor rather than maximum leaderboard performance.

A ten-run bootstrap stability analysis on the training set showed Brier score standard deviations below 0.001 for all three model classes, confirming that the model comparison was stable rather than a single-split artifact. On the held-out post-2016 test set, Brier score differences between logistic regression and the boosting alternatives were at most 0.0012 (Table I). Logistic regression with isotonic calibration was selected as the decision-support model for its transparent coefficient structure, stable training behavior, and negligible performance difference from more complex alternatives.

Calibration was applied after fitting using isotonic regression on a held-out calibration split (126,135 pre-2016 loans), following the prefit calibration procedure to avoid refitting the base model [14]. On the post-2016 test set, isotonic calibration produces $ECE = 0.033$. Calibration quality varies by difficulty tier (easy: 0.025; medium: 0.040; hard: 0.050), reflecting higher uncertainty in the ambiguous mid-range where the model has less predictive leverage.

C. Case Selection and Difficulty Stratification

Eighteen scored cases and two practice cases were used. Case difficulty was defined model-independently using empirical default rates estimated on training data, avoiding circularity between difficulty and model uncertainty. For each combination of interest-rate, DTI, and income bin (3D grid),

TABLE II
INTERACTION PROTOCOLS USED IN THE WITHIN-SUBJECT EXPERIMENT

Protocol	Participant flow
No AI	Loan profile; final probability estimate; approve/reject decision.
AI-first	Loan profile with AI probability and recommendation; final probability estimate; approve/reject decision.
Human-first	Loan profile; initial estimate and decision; AI probability and recommendation; final estimate and decision.

a Bayesian-smoothed default rate was computed, shrinking sparse-bin estimates toward the training mean:

$$\hat{p}_b^{\text{smooth}} = \frac{n_b \bar{p}_b + \alpha \bar{p}_{\text{train}}}{n_b + \alpha}, \quad \alpha = 50. \quad (1)$$

Difficulty was then defined as closeness to maximum uncertainty:

$$\text{difficulty}_i = 1 - \frac{|\hat{p}_i - 0.5|}{0.5}, \quad (2)$$

yielding a score in $[0, 1]$ partitioned into three tiers (easy, medium, hard) at the 33rd and 66th percentiles. Smoothing sensitivity analysis confirmed that tier assignments were stable for $\alpha \in [10, 100]$.

Scored cases were sampled from the post-2016 test set to span 3 difficulty tiers $\times$ 2 model-correctness states = 6 cells, with 3 cases per cell, for a total of 18. Within each cell, cases were selected to maximize diversity of predicted probability (minimum pairwise gap of 0.02) and loan purpose. The selected cases span predicted probabilities from 0.055 to 0.946 (mean 0.441). All case files were locked before participant recruitment and have not been modified since.

D. Interaction Protocols

The experiment used three within-subject protocols. In the *no-AI* protocol, only the loan profile was displayed; participants entered a final binary decision and a probability estimate. In the *AI-first* protocol, the model’s predicted default probability and a derived cost-sensitive recommendation were displayed before the participant’s decision. In the *human-first* protocol, participants first submitted an initial binary decision and probability estimate; the AI probability and recommendation were then revealed; participants then had the opportunity to revise both responses before final submission.

Table II summarizes the three protocol structures. The human-first protocol is the only condition that enables pre/post revision measurement and WOA analysis; the initial and final submissions are stored separately in the database.

E. Experiment Platform and Reproducibility

The experiment was implemented as a web application in Streamlit with a Supabase backend. The platform enforced protocol-specific trial flow, stored participant records, trial-level decisions, probability estimates, protocol labels, case identifiers, and timing metadata at the individual trial level.

This implementation allows behavioral data to be reconstructed and linked to frozen experimental cases without storing personally identifying information.

Reproducibility was handled through locked artifacts. The case-design notebook produced candidate pools and working outputs to a build directory; the official experimental snapshot was stored separately as frozen case files with a lock file (`cases.lock.json`) and integrity validation. The submitted repository separates the official behavioral reproduction from the optional upstream raw-data rebuild. The command `python run.py` regenerates the final reported tables, figures, and summary from `data/experiment_exports/` and `artifacts/frozen/`. All paper figures are generated programmatically by the reproduction pipeline and copied into the paper source without manual editing. The raw file `data/raw/loan.csv` is needed only for the optional provenance command `python run.py --mode rebuild`, which writes to `artifacts/build/` and does not modify the frozen experimental boundary.

F. Participants

A total of 158 participants enrolled; 100 completed all scored trials (63.3% completion rate). A chi-square balance test confirmed that the three counterbalancing groups were not meaningfully unbalanced among completers ($\chi^2 = 0.560$, $p = 0.756$): 30 in group 1, 36 in group 2, and 34 in group 3. Each completed participant contributed 18 scored trials, yielding 1,800 scored decisions in total.

The sample was predominantly young (78% aged 18–24; 14% aged 25–34) and student-based (56% bachelor’s in progress; 28% master’s or higher). No participant reported professional lending experience. Participants completed a three-question comprehension quiz before scored trials; 80.2% passed on the first attempt.

G. Outcome Measures and Statistical Analysis

The primary outcome was mean trial cost under the asymmetric cost function:

$$\text{cost}(d, y) = \begin{cases} C_{FN} = 5 & \text{if } d = \text{approve}, y = \text{default} \\ C_{FP} = 1 & \text{if } d = \text{reject}, y = \text{paid} \\ 0 & \text{otherwise,} \end{cases} \quad (3)$$

where d is the participant’s binary decision and y is the true loan outcome. This ratio implies the normative cost-minimizing threshold $\tau = C_{FP}/(C_{FP} + C_{FN}) = 1/6 \approx 0.167$ [15]: approve if and only if predicted default probability is below τ . Accuracy was a secondary outcome.

Weight of advice was computed for human-first trials as:

$$\text{WOA}_i = \frac{p_i^{\text{final}} - p_i^{\text{initial}}}{p_i^{\text{AI}} - p_i^{\text{initial}}}. \quad (4)$$

Trials where $|p_i^{\text{AI}} - p_i^{\text{initial}}| < 0.01$ were excluded (21 of 600; 3.5%) as adjustment toward the AI estimate was undefined in such cases. Values were clipped to $[-1, 2]$ for stability.

The primary inference used paired t -tests on per-participant protocol means, which directly follows from the within-subject

TABLE III
PRIMARY OUTCOME: MEAN TRIAL COST BY PROTOCOL ($N = 100$)

Comparison	Mean A	Mean B	Δ	p
No AI vs. AI-first	1.222	1.012	+0.210	0.017
No AI vs. Human-first	1.222	0.923	+0.298	< 0.001
AI-first vs. Human-first	1.012	0.923	+0.088	0.178
Model-optimal ($\tau = 1/6$) [†]		0.333	—	—

Δ = Mean A – Mean B; positive = A is worse.

[†]Observed realized cost from applying $\tau = 1/6$ to AI probabilities.

95% CIs: no AI vs. AI-first (0.038, 0.382); no AI vs. HF (0.144, 0.452).

TABLE IV
SECONDARY OUTCOME: ACCURACY BY PROTOCOL ($N = 100$)

Comparison	Mean A	Mean B	Δ	p
No AI vs. AI-first	0.552	0.635	−0.083	< 0.001
No AI vs. Human-first	0.552	0.650	−0.098	< 0.001
AI-first vs. Human-first	0.635	0.650	−0.015	0.408

Δ = Mean A – Mean B; negative = A is less accurate.

Effect sizes: no AI vs. AF $d = -0.392$; no AI vs. HF $d = -0.467$.

design and avoids pseudo-replication from treating 1,800 trials as independent. Three planned contrasts were evaluated for each outcome: no AI vs. AI-first, no AI vs. human-first, and AI-first vs. human-first. A trial-level linear mixed model was attempted as a robustness check but produced a singular covariance matrix, indicating the random-effects structure was over-parameterized for the within-participant variance structure; paired t -tests are therefore the sole primary inference. Effect sizes are reported as Cohen’s d .

IV. RESULTS

A. Decision Cost

Table III shows the primary result. Mean trial cost was 1.222 in the no-AI condition, 1.012 with AI-first, and 0.923 with human-first. Both AI-assisted protocols significantly reduced cost relative to no AI. Human-first produced the lowest cost numerically. The direct timing contrast between AI-first and human-first was not statistically decisive: $\Delta = 0.088$ (95% CI: $[-0.041, 0.218]$; $t = 1.356$, $p = 0.178$, $d = 0.136$).

B. Decision Accuracy

Table IV shows accuracy contrasts. Mean accuracy was 55.2% without AI, 63.5% with AI-first, and 65.0% with human-first. Both AI-assisted protocols significantly improved accuracy relative to no AI; the AI-first vs. human-first contrast was not significant.

The AI benefit by difficulty tier (Table V) reveals that the improvement is concentrated in easy cases: both AI protocols improved accuracy by approximately 22 percentage points over no AI for easy cases but by at most 4–5 percentage points for medium and hard cases. Because difficulty was defined using training-data empirical default uncertainty rather than participant-rated difficulty, these tiers should be interpreted as statistical case strata rather than subjective difficulty ratings.

TABLE V
ACCURACY BY PROTOCOL AND DIFFICULTY TIER

Difficulty	No AI	AI-first	Human-first
Easy	59.0%	81.0%	82.0%
Medium	58.0%	58.5%	62.5%
Hard	48.5%	51.0%	50.5%

200 trials per cell.

Protocol-level outcomes

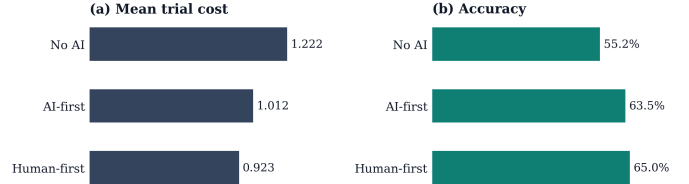


Fig. 1. Mean trial cost (left) and accuracy (right) by protocol across 100 participants. Lower cost and higher accuracy are better. Confidence intervals for all pairwise contrasts are reported in Tables III and IV.

The protocol-level cost and accuracy pattern is summarized in Fig. 1.

C. Human-First Correction Behavior

Across 600 human-first trials, 300 (50.0%) stayed correct after AI exposure, 90 (15.0%) improved from initially wrong to finally correct, 178 (29.7%) stayed wrong, and 32 (5.3%) worsened from initially correct to finally wrong (Table VI). The net gain was +58 trials (9.7 percentage points). Among the 122 trials where correctness changed, a one-sided exact binomial test confirmed that improvements significantly outnumbered worsenings (90 vs. 32; $p < 0.0001$).

The same pre/post correction structure is visualized in Fig. 2.

Human-first correction matrix

	Final correct	Final wrong
Initial correct	300 Stayed correct	32 Worsened
Initial wrong	90 Improved	178 Stayed wrong

Fig. 2. Human-first correction matrix. Green corresponds to trials that improved after AI exposure, red to trials that worsened, and gray to trials whose correctness status did not change.

TABLE VI
HUMAN-FIRST PRE/POST CORRECTION MATRIX ($n = 600$ TRIALS)

	Final correct	Final wrong
Initial correct	300 (50.0%)	32 (5.3%)
Initial wrong	90 (15.0%)	178 (29.7%)

D. Weight of Advice and Reliance Heterogeneity

Of 579 WOA-eligible human-first trials, 304 (52.5%) showed essentially no probability movement (absolute change < 0.01). Among the 275 adjusted trials, median WOA was 0.869 and mean WOA was 0.861 (Fig. 3). The overall WOA distribution (mean 0.409; median 0.000) conceals a bimodal pattern: non-adjustment versus near-complete adoption of the AI estimate among those who did adjust.

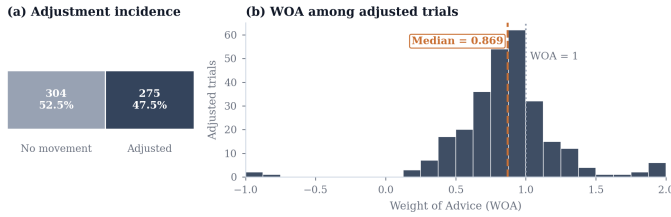


Fig. 3. (a) Adjustment incidence among 579 WOA-eligible human-first trials. (b) WOA distribution among the 275 adjusted trials (those with $|p^{\text{final}} - p^{\text{initial}}| \geq 0.01$). Dashed line: median WOA = 0.869; dotted line: WOA = 1 (full adoption of AI estimate).

Individual-level AI benefit on accuracy (mean = 0.091, SD = 0.191) further confirms substantial between-participant heterogeneity. The distribution has a negative tail: approximately one-third of participants showed lower accuracy under AI-supported protocols than under no AI, suggesting a subgroup for whom AI support was not beneficial.

V. ANALYSIS AND VALIDATION

A. Did the Study Objectives Succeed?

The study had three objectives: (1) quantify the effect of AI support on cost-sensitive decisions; (2) test whether timing of AI advice modulates decision quality; (3) characterize reliance behavior in the human-first condition. On objective 1, both AI protocols reduced cost and improved accuracy with medium effect sizes, and the human-first correction analysis confirmed net-positive AI-induced corrections ($p < 0.0001$, binomial). On objective 2, the timing contrast was directional but not statistically decisive at $N = 100$ ($p = 0.178$; 95% CI includes zero). On objective 3, WOA analysis documented strongly bimodal reliance, and the correction matrix decomposed AI-induced changes into beneficial and harmful revisions.

B. Alignment with Prior Work

The magnitude of AI benefit aligns with prior algorithm-in-the-loop findings that decision support can improve unaided human judgment but rarely realizes the full standalone model potential [6], [10]. The concentration of benefit in easy cases mirrors Bansal et al.'s observation [10] that complementary

TABLE VII
COST BENCHMARKS ON THE 18-CASE SET

Strategy	Cost
Always approve	2.500
Random (50%)	1.944
Model at $p < 0.5$	1.833
No AI (observed)	1.222
AI-first (observed)	1.012
Human-first (observed)	0.923
Always reject	0.500
Model-optimal ($\tau = 1/6$, observed)	0.333

human-AI performance depends on task structure: where the AI is confident and correct, the benefit is large; where the AI is ambiguous, humans provide limited additional signal.

The human-first correction behavior is consistent with the cognitive forcing function literature [2]. Requiring initial judgment before AI exposure produced measurably more corrections than degradations (90 : 32). However, 178 trials (29.7%) remained wrong even after AI exposure, indicating that the forcing function alone is insufficient for cases where participants lacked the domain knowledge to integrate the AI signal correctly.

The WOA distribution is consistent with discrete switching behavior documented in advice-taking research [11], [12]. The bimodal pattern—ignoring versus fully adopting—is less consistent with the conservative-updating framework, which predicts intermediate WOA values. This suggests that participants in this domain treated AI advice as a categorical override signal rather than a Bayesian update.

C. Comparison with Benchmark Strategies

Table VII compares observed human protocol means with reference decision strategies on the 18-case set. Human performance under all protocols substantially exceeded the always-approve and random baselines, indicating that participants were genuinely discriminating. However, all human conditions fell substantially above the model-optimal benchmark (0.333), a gap that even human-first support (0.923) did not close. This gap reflects both imperfect calibration and sub-optimal threshold application, not model quality.

D. Robustness Checks

Group-3 sensitivity. Excluding the 34 participants in counterbalancing group 3 (reduced sample $N = 66$) preserved all substantive conclusions. The no AI vs. AI-first contrast remained significant ($t = -4.959$, $p < 0.0001$, $d = -0.610$) and no AI vs. human-first remained significant ($p < 0.0001$, $d = -0.660$). The AI-first vs. human-first contrast remained non-significant ($p = 0.658$).

Non-parametric check. Wilcoxon signed-rank tests on per-participant accuracy means produced the same pattern of significant and non-significant contrasts as the paired t -tests, indicating that the results are not driven by distributional assumptions.

Model selection stability. Bootstrap resampling over 10 runs confirmed Brier score standard deviations below 0.001 across all three candidate models. The final logistic regression model was selected before any participant data were collected; model evaluation on the post-2016 test set served as a final reporting diagnostic only.

E. Normative Deviation and Threshold Analysis

Normative deviation from the cost-sensitive threshold was 26.0% for easy cases, 73.3% for medium cases, and 4.7% for hard cases. The medium tier is the region where the optimal rule ($\tau = 1/6$) requires rejection but a naive accuracy-maximizing rule at 0.5 would approve. Participants deviated most frequently in this region, consistent with application of a 0.5 threshold rather than the cost-calibrated τ .

Implicit decision threshold estimates, derived by fitting logistic regression to no-AI trial decisions, yielded valid estimates for 96 of 100 participants (mean = 0.355, median = 0.299), substantially above the normative $\tau = 0.167$. This confirms that participants systematically approved loans at predicted default probabilities that exceed the cost-minimizing threshold, contributing to the persistent gap from the model-optimal benchmark.

Decision consistency between final binary decisions and probability estimates was 84.6% at $\tau = 0.167$ and 79.1% at $\tau = 0.5$, indicating that participants' own probability estimates were more consistent with the cost-sensitive rule than their actual decisions. The case-level cost pattern is shown in Fig. 4.

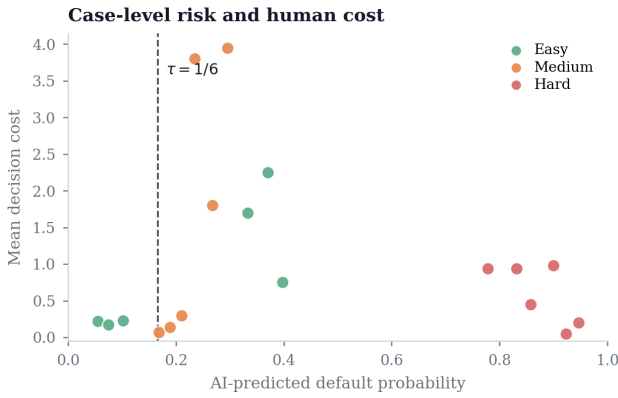


Fig. 4. Case-level mean cost by AI-predicted default probability. Each point is one of the 18 scored cases aggregated across participants; the dashed vertical line marks the cost-sensitive threshold $\tau = 1/6$.

F. Validation Limits

The validation evidence has three main limits: the sample was young and non-expert, each participant completed only 18 scored trials, and the mixed-effects model did not converge. Therefore, the results should be read as controlled evidence about interaction design, with the fuller generalizability and power limits discussed in Section VI.

VI. DISCUSSION AND CONCLUSION

A. Main Contributions

This paper provides three empirical contributions to human-centered AI research. First, a within-subject experiment demonstrating that the same calibrated model improves cost-sensitive decisions under both AI-first and human-first protocols, with the timing contrast directional but not statistically decisive. Second, a decomposition of AI-induced revision behavior showing that human-first exposure corrects more errors than it introduces. Third, documentation of bimodal WOA behavior in a financial domain with explicit cost asymmetry.

B. Design Implications

The system-level framing has direct implications for AI deployment. In a practical screening system, low-risk and high-confidence cases could be routed through an AI-first workflow to reduce user burden, while borderline or high-stakes cases could require a human-first workflow in which the user records an independent judgment before seeing the model output. This preserves independent deliberation where anchoring is most costly while still allowing efficient support where the model signal is strong. The cost gap between no AI (1.222) and model-optimal (0.333) shows that model quality is a necessary but insufficient condition for decision improvement. Interaction structure mediates how much of that potential is realized. AI-first designs are efficient and show measurable improvement, but risk anchoring users on the model before they have formed an independent view. Human-first designs preserve deliberation and enable revision analysis, but create inertia and require more interface complexity.

A practical system might apply the human-first structure for high-stakes or ambiguous cases—where independent deliberation has the most value—while allowing AI-first for cases where the model is highly confident, an idea related to learnable deferral [7]. Additionally, results suggest that probability support should be paired with explicit decision-threshold guidance, as the normative gap between the cost-sensitive τ and the naive 0.5 threshold was a primary contributor to suboptimal decisions. This aligns with Zhang et al.'s [13] finding that calibrated confidence improves calibration only when users know how to interpret it.

C. Limitations

Several constraints bound interpretation. The sample consisted mainly of young, non-expert participants (78% aged 18–24, no professional lending experience), limiting generalization to loan officers or credit analysts. The experiment used 18 scored trials per participant, which limited statistical power for detecting the small AI-first vs. human-first effect ($d = 0.136$); a simulation-based power analysis [16] suggests $N \approx 350$ –450 would be required for 80% power at the observed effect size. The linear mixed model failed to converge, precluding block-order adjustment; though paired tests are robust, learning effects within session cannot be formally separated from protocol effects. The AI advice is a scalar probability without

feature attribution; results may differ under explainable AI paradigms [13] that provide feature-level rationales.

Internal validity is strengthened by the within-subject design, counterbalanced protocol order, locked case set, and fixed AI model, but it is not complete. Because each participant experienced all three protocols in a single session, practice and fatigue effects may remain partially confounded with protocol exposure despite counterbalancing. Construct validity is also bounded: the task captures structured probability-assisted decision making, not the full institutional, legal, and ethical context of real credit allocation. External validity is therefore limited to similar decision-support settings where users interpret probabilistic advice under explicit asymmetric costs.

D. Future Directions

Future work should examine the timing contrast with larger and more expert samples, more trials per participant, and populations with professional decision-making experience. The WOA polarization pattern warrants investigation into individual-level moderators—such as domain knowledge, numeracy, and prior AI exposure—that predict whether a participant will adjust or anchor. Hybrid workflow designs that adaptively apply human-first or AI-first structure based on case difficulty or model confidence represent a promising applied direction [7]. Finally, the persistent gap from the model-optimal benchmark across all human conditions suggests that threshold education and cost-framing interventions deserve investigation as complements to advice-timing manipulations.

E. Conclusion

This paper evaluated AI advice timing in a controlled cost-sensitive decision-support experiment. One hundred participants completed a within-subject task comparing no AI, AI-first, and human-first protocols. AI support improved decision quality relative to no AI under both cost and accuracy metrics. Human-first produced the strongest descriptive performance and corrected more initial errors than it introduced, but the direct timing difference between human-first and AI-first was not statistically decisive. The main conclusion is that human-AI decision quality is a property of the whole interaction system: a calibrated model can improve decisions, but the realized gain depends on when advice is shown, whether users revise appropriately, and whether decisions are evaluated against the correct cost structure. For human-centered AI applications requiring high-stakes decision support, model validation should therefore be paired with behavioral validation of the interaction workflow through which model outputs reach human judgment.

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APPENDIX A EXPERIMENT PLATFORM IMPLEMENTATION

The behavioral experiment was implemented as a Streamlit web application with Supabase/PostgreSQL persistence. The participant-facing app used a phase-router structure: the entry point established a browser session identifier, resumed incomplete sessions when the same identifier was found in Supabase, routed completed sessions to an already-completed screen, and selected the appropriate UI screen from a phase-to-screen router. The main participant flow was consent, demographics, glossary and instructions, comprehension quiz, two practice trials with feedback, three scored blocks of six trials each, reflection questions, trust rating, performance summary, and thank-you screen.

Each scored block used exactly one protocol (`no_ai`, `ai_first`, or `human_first`), and each participant saw all three protocols exactly once. Protocol order was assigned using the locked Latin-square rotation for `group_1`, `group_2`, and `group_3`. The deployed platform loaded the 18 scored cases, two practice cases, and protocol rotation from frozen CSV files rather than manually embedding experimental stimuli in code.

The Supabase schema used three principal tables: `participants`, `trials`, and `quiz_responses`. The `participants` table stored group assignment, completion state, session-resumption fields, total and optimal cost, trust rating, and post-trial reflection responses. The `trials` table stored one row per practice or scored trial, including protocol, case identifier, AI prediction, true outcome, initial/final judgments, probability estimates, timing fields, and correctness indicators. The `quiz_responses` table logged comprehension-quiz answers by participant, attempt, and question. Trial rows used a uniqueness constraint on (`participant_id`, `trial_index`) to prevent duplicate submissions.

The implementation separated responsibilities across modules: the app entry point handled session bootstrap and routing, a database module isolated Supabase reads/writes, and an experiment-logic module handled group assignment, deterministic trial sequencing, and scoring. A demo mode was available for presentation without storing study responses. After data collection, completed participant, trial, and quiz tables were exported from Supabase as CSV files under `data/experiment_exports/`, which are the direct inputs to the official reproduction pipeline.

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